

Population Age structure

Introduction:

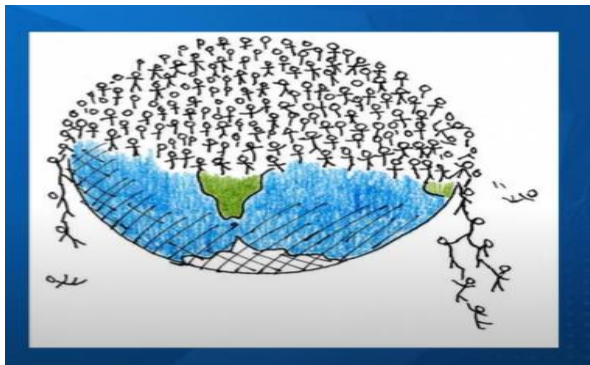
The **age structure** of a population refers to the proportionate numbers of people in different age categories in a given population for a **defined time**. It is a natural characteristic of a population in a country or a region. The age structure is closely related to the birth rate, death rate and migration of a population

1. Few concepts are needed to understand the population age structure

1.1 Total fertility rate: The average number of children that would be born to a woman in her lifetime if the age-specific birth rates remain constant.

- In developed nations TFR: 1.9
- In developing nations TFR: 4.7
- In 1960's in India TFR was ~6
- It is 2.68 in 2007, 2.40 in 2015, and **2.159 in 2022**

1.2 The earth's carrying capacity:



Carrying capacity: The number of individuals who can be supported in a given area within natural resource limits, and without degrading the natural, social-cultural, and economic environment for present and future generations.

- If the carrying capacity is exceeded, it will negatively impact sustainable development
- No population can live beyond the environment's carrying capacity for a long time

- A study indicates the earth's carrying capacity as 10 billion people.

1.3. Exponential growth:

- When any quantity increases by a constant amount per unit time, 1,3,5,7 etc., it is called linear growth. However, when it increases by a fixed percentage, it is called exponential growth. This exponential growth is the reason for the enormous rise in global population.

1.4. Doubling time:

The time needed for a population to double its size at a constant annual rate is called doubling time. It is calculated as follows:

Td = $70/r$, where

Td = doubling time in years

r = annual growth rate.

If a nation has a 2% annual growth rate, its population will double in 35 years.

1.5. Population size: is the number of individuals present in the country

1.6. Mortality: is the number of deaths of individuals per unit time.

1.7 Zero population growth (ZPG)

When birth plus immigration in a population is just equal to deaths plus emigration, it is said to be zero population growth



1.8. Population profile: It is also called **age structure**. It is a bar graph plotted with number of males and females for successive ages in the population, with the oldest or the aged at the top of the pyramid. The males are represented on the left hand side in blue, while the females are represented on the right using a pink colour..

It is common in demography to split the population into three broad age groups:

Stage1: **pre-reproductive stage** 0-14 years, children and young adolescents

Stage2: **reproductive stage** 15-60 years, the working-age population

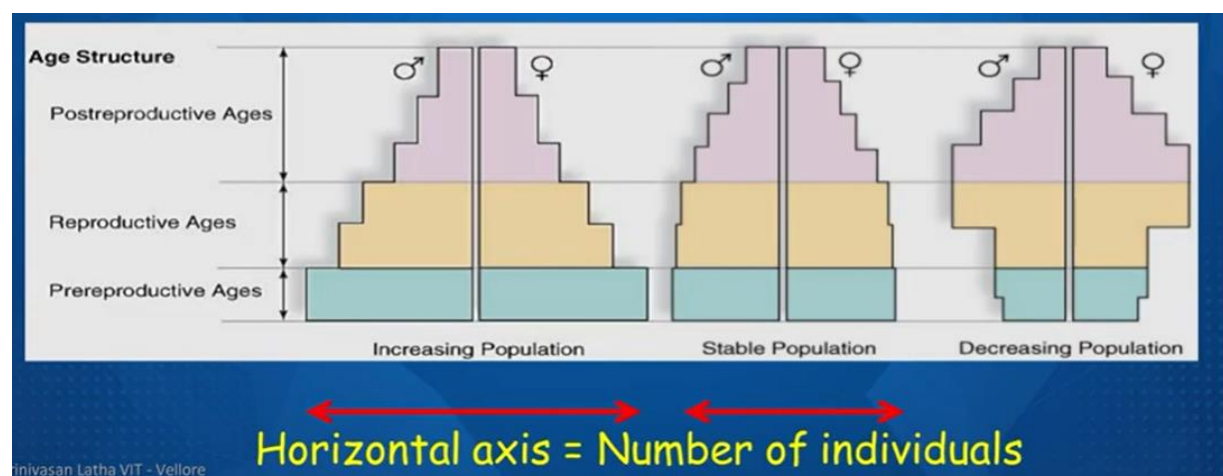
Stage3: **post-reproductive stage** 60 or 65 plus, the elderly population

Note that:

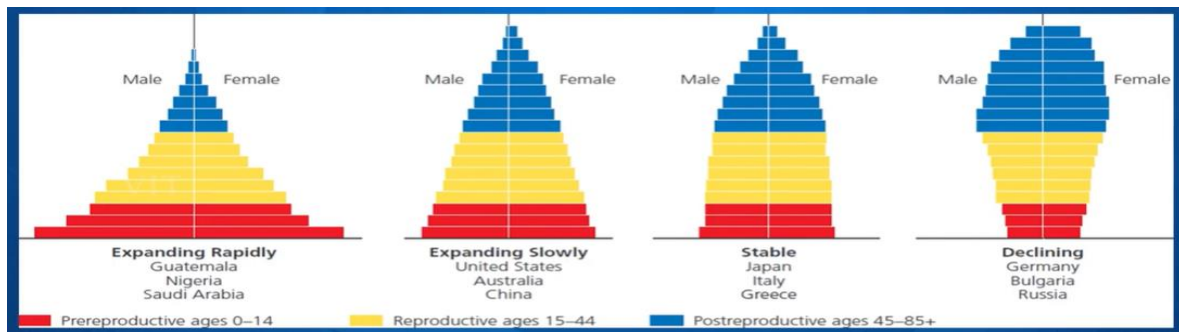
(In some demographic graphs age groups are grouped as 0-14; 15-44 and 45 and above years)

2.Age Structure and their types:

Different types pyramid, bell, urn shape, population growth can be predicted. Here horizontal axis is the number of individuals and the vertical axis represents age structure. Age structure is divided into three stages: pre-reproductive ages, reproductive age, and post-reproductive age.



Here are some more examples of countries with expanding rapidly, expanding slowly, stable, and declining population age structures.



In these images, we can see the shape of the population according to the three stages discussed earlier: depending on pre-reproductive, productive, and post-reproductive stages. In these images, males are represented on the left hand and females on the right hand.

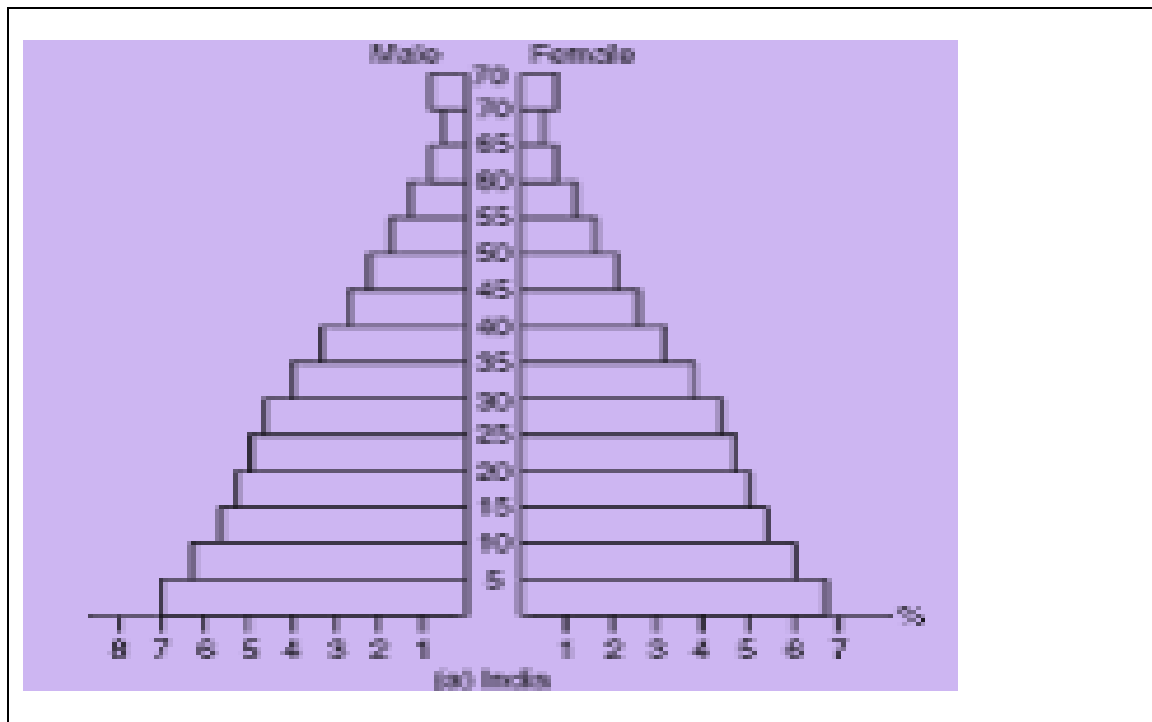
2.1 Pyramid shaped:

Here the very young population is more, making a broad base and old people are less. This type indicates **growing** population.

- The large number of individuals in very young age will soon enter into reproductive age, thus causing an increase in population, whereas
- Less number of people in old age indicate less loss of population due to death.

- India, Bangladesh, Ethiopia, Nigeria are examples of this type

Case 1: increasing population: A progressive age structure is one in which both birth and death rates are high. Here, wide base of pre-reproductive individuals, indicating a high fertility rate and growing population. Children account for only 45-55% of the total population and the aged for only 5-10%. Such a structure is common in developing countries where social, cultural, and perhaps religious and economic conditions lead to high fertility, and poor living conditions, bad diets, and little medical aid lead to high levels of mortality.

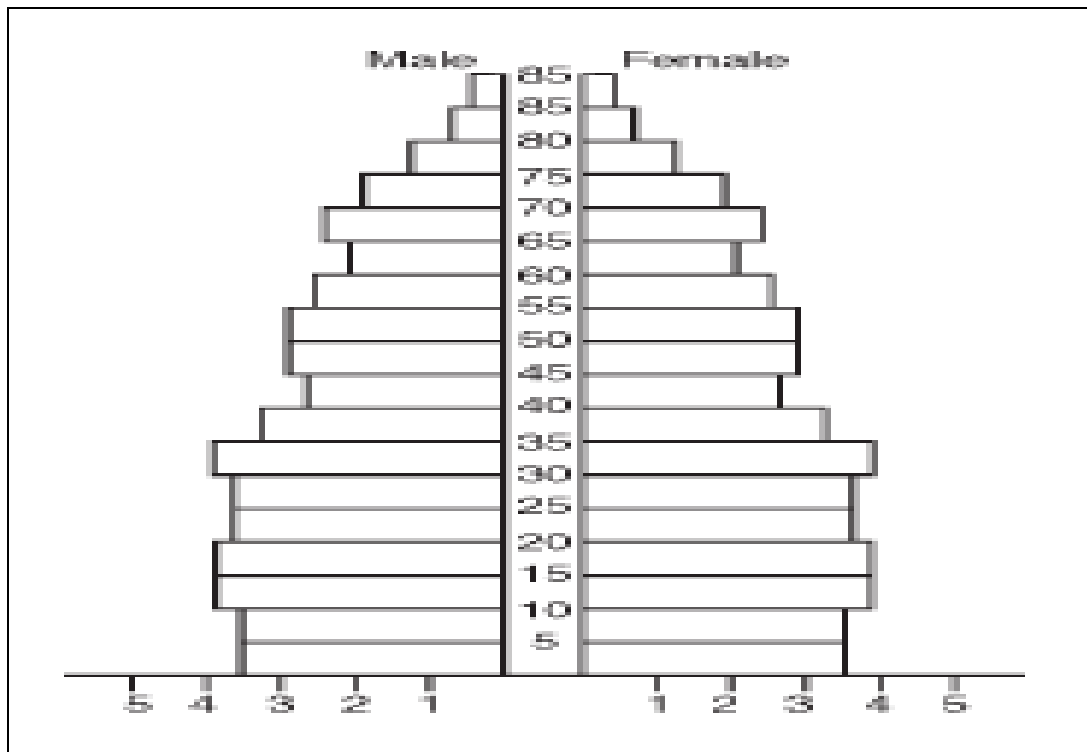


India as an illustration: Source courtesy: Textbook: Environmental Science and Engineering, Kaushik and Kaushik, New age publishers.

2.2 Bell shaped:

It occurs in countries like France, USA and Canada where birth rates have in the past one or two decades declined resulting in people of almost equal number in age group 0-35 years.

So in the next 10 years, the people entering into reproductive age group is not going to change much and such age-pyramids indicate **stable** populations.



France as an illustration: Source courtesy: Textbook: Environmental Science and Engineering, Kaushik and Kaushik, New age publishers.

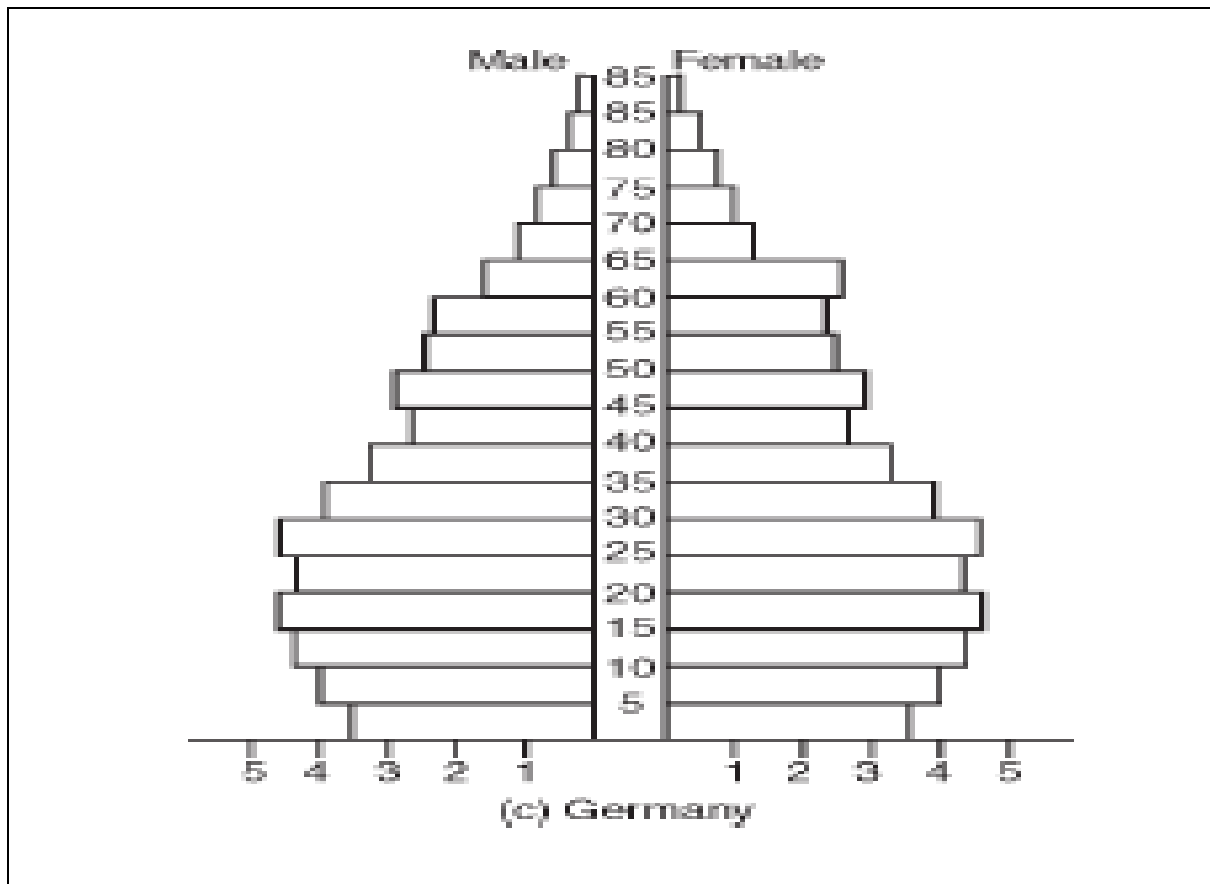
Case 2: stable population: A stationary age structure is one in which birth and death rates are both low and children account for about 35-40% of the total population and the aged for about 10%. Here, Pre-reproductive and reproductive groups have similar widths, indicating a fertility rate at about replacement level and results in a stable population. This pattern may remain the same for many years

2.3 Urn shaped:

Here number of individuals in very young class is smaller than the middle reproductive age class.

In the next 10 years the number in reproductive age class will thus become less before resulting in a **decline** of population.

Germany, Italy, Hungary, Sweden and Japan are examples of this type.



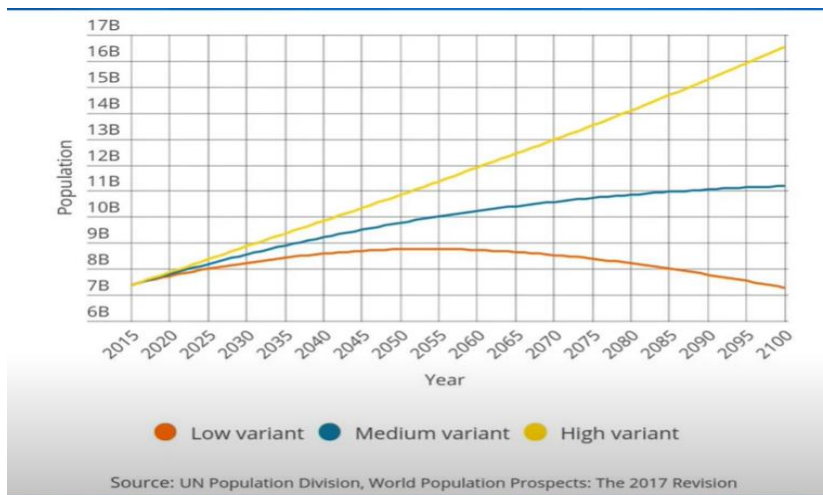
Germany as an illustration: Source courtesy: Textbook: Environmental Science and Engineering, Kaushik and Kaushik, New age publishers.

Case 3: Decreasing population: A regressive age structure is one in which birth and death rates are low and declining. Here, pre-reproductive age groups are narrower than reproductive age groups, indicating a low fertility rate and resulting in a shrinking population. Children account for about 30% of the total population and the aged for above 15%. This pattern is common in developed countries (especially those in Western Europe), where high living standards, education, and social awareness are accompanied by good food and medicine.

3.India population:

The present population of India is 1.21 billion and these numbers are increasing every day. From the figure shown below, it can be concluded that by 2050 women will have an average

of 2.5 children which is the highest estimate to the lowest estimate of 1.5 childre



4. Effect of growing population on the nature



Reduction of Biodiversity



Increasing genetic resistance of pest species and disease-causing bacteria



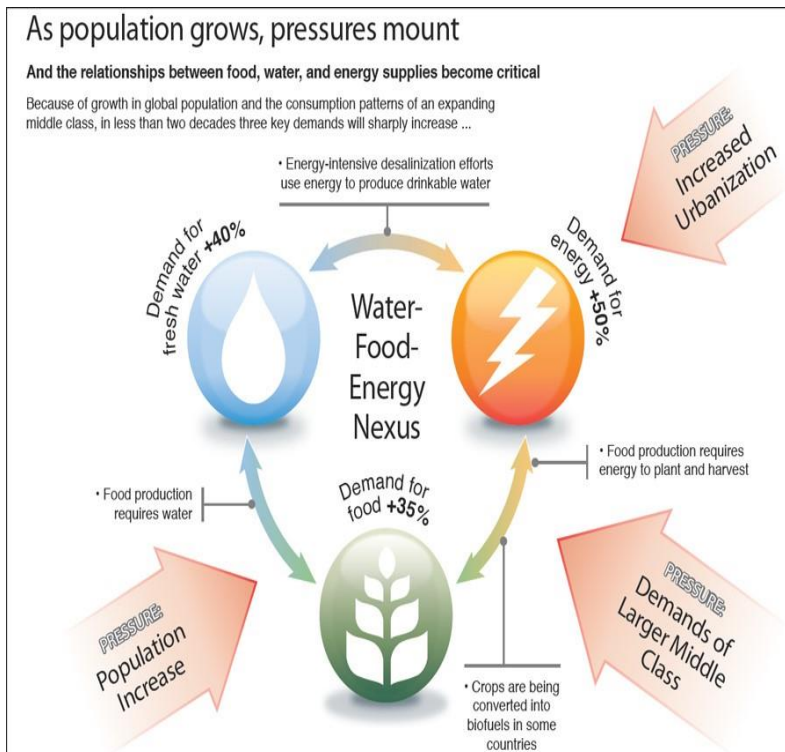
Using some renewable resources faster than they can be replenished



Interfering with the earth's chemical cycling and energy flow processes



Relying mostly on polluting and climate-changing fossil fuels



source courtesy: <https://www.blendspace.com/lessons/SzQzdtRv3gVK5Q/population-growth-and-natural-resources>

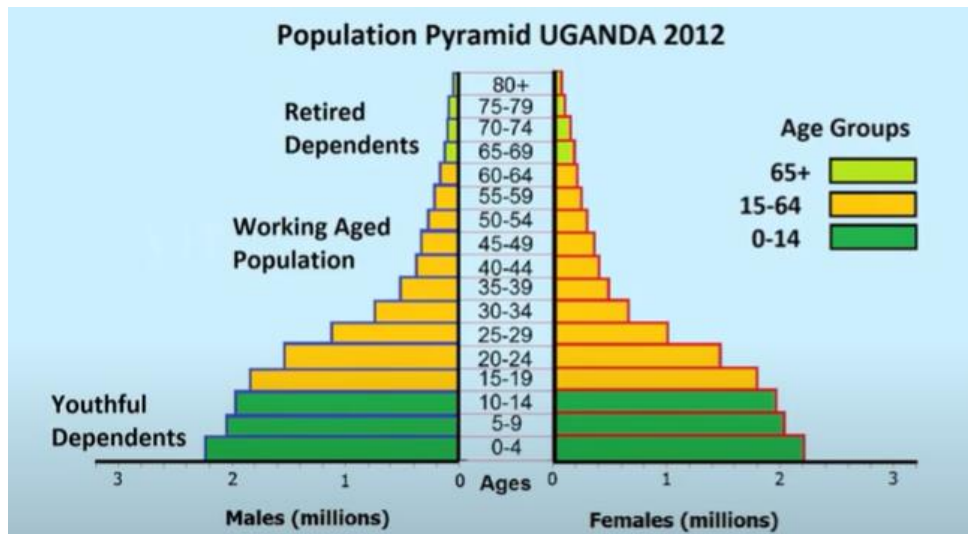
5. Factors that are influenced by the age structure:

The age structure of a population affects a nation's key socioeconomic issues.

- Countries with young populations (high percentage under age 15) need to invest more in schools
- Countries with older populations (high percentage ages 65 and over) need to invest more in the health sector.
- The age structure can also be used to help predict potential political issues. For example, the rapid growth of a young adult population unable to find employment can lead to unrest

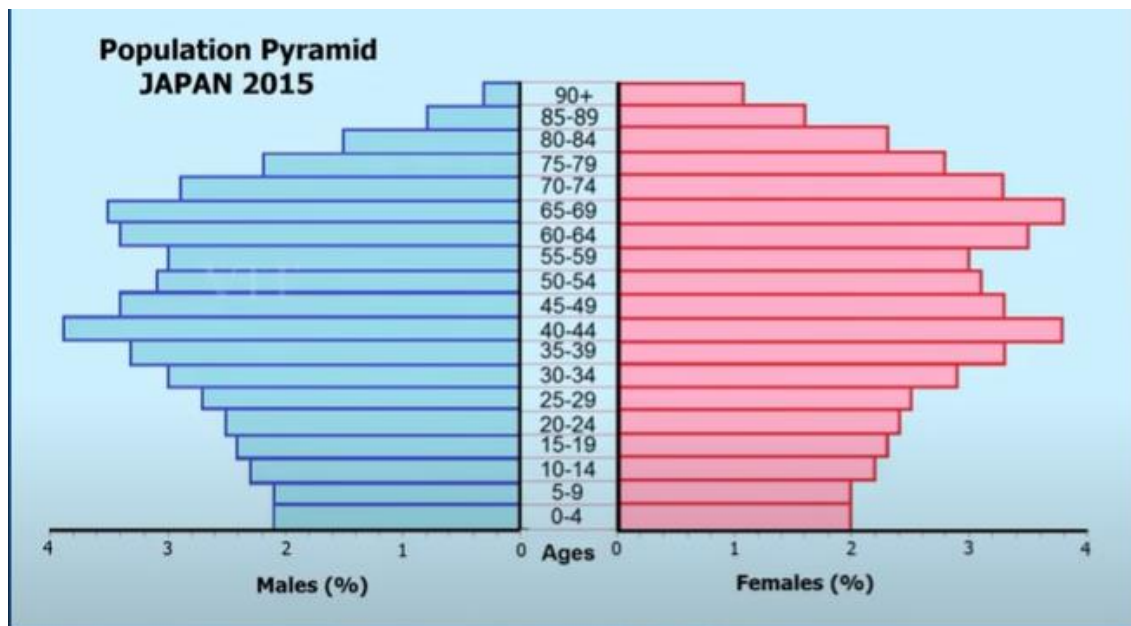
6. Interpretation of age structure using examples:

Example 1: Population pyramid of UGANDA 2012



- i. As a result of lower mortality but still high fertility, Uganda has developed a very youthful age structure.
- ii. Uganda's population will continue to grow because of the large number of people who are either currently at an age when they are having children or who will soon enter that age group.
- iii. More youthful dependence means the government has to give more money and adapt schemes for child welfare, schools and their policies.
- iv. These policies or schemes are important to implement considering they are going to be the next reproductive stage. T
- v. herefore, government has to concentrate on education, health and psychological impacts considering that these children who will be adults in the future impact the country.
- vi. So, UGANDA is progressing as these children grow up provided, they are given good opportunities for education, better resources for upskilling and will also be productive to the nation in the future.

Example 2: Population pyramid of JAPAN 2015



The above figure is the population age structure in Japan in 2015, where we can see a big urn-like structure. We see the male and female congregation and also people living up to 90 plus age. Also, we see a smaller number of pre-reproductive populations i.e., a smaller number of children whether it is male or female. Therefore, a large number of populations are in the middle group / middle band, which is working for the band and making sufficient money for their living. Also, the nation can use the talent and convert it into a progressive ratio of the country based on this talent pool which is available with the working population. Here, government can provide training opportunities and nurturing of their population's talent to convert into productive outcomes. Here government focus is on training process, workplace ethics and etiquette, targeted skills, and skill training structures for a particular band. However, there is a hitch in this graph as the life expectancy of the Japanese is pretty high, which means the elderly persons at home. This means the government has to spend lots of money on medical facilities to provide health care facilities and also to the ones who are taking care of elderly people and address their emotional physical and psychological needs. Because of the lots of development in the healthcare systems, and medical transformations, peoples have much better lifestyles.

How to slow down population

- Promote economic development



- Promote-family-planning



- Promote women empowerment



7. Summary:

We learned about population age structure, characteristics of population age structure, how to draw it and depiction, understanding the number of male and female age structure in the groups and its impact on the population size. We also learned about the definitions of the fertility rate, doubling time, and zero population growth.

